Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: _____

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

□ Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (□ Easy)

Description:
(Use this space to write a description of the scenario in 4–6 lines.)----------

Task	Description
1.	Load the `ign.csv` dataset using Pandas.
2.	Display the first 5 rows (`head()`), last 5 rows (`tail()`), and the shape of the dataset.
3.	Remove the unnecessary column: `"Unnamed: 0"`.
4.	Check for missing values in `score`, `genre`, and `platform`.
5.	Handle missing values: Fill the numeric column `score` with the mean, and the categorical column `genre` with the mode.
6.	Ensure correct data types: Convert `score` → float and `release_year`, `release_month`, `release_day` → integer.

Graph (Optional):

(Describe the graph and mention the graphs here)
(Insert graph here if applicable else remove the Graph heading and the graph info)

Conclusion:
(Use this space to write the conclusion for the scenario in 4–6 lines.)----------

Scenario 2: Line Graph (Score Trend) + Save (□ Easy)

Description:
(Use this space to write a description of the scenario in 4–6 lines.)----------

Task	Description
1.	Group data by `release_year`.
2.	Calculate the average score per year using Pandas.
3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with `release_year` on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: `plt.savefig("avg_score_trend.png")`.

Graph (Optional):
(Describe the graph and mention the graphs here)
(Insert graph here if applicable else remove the Graph heading and the graph info)

Conclusion:
(Use this space to write the conclusion for the scenario in 4–6 lines.)----------

Scenario 3: Filtering + Bar Chart + Save (□ Medium)

Description:
(Use this space to write a description of the scenario in 4–6 lines.)----------

Task	Description
1.	Filter the dataset to include only games where `score > 7`.
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with `platform` on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: `plt.savefig("top_platforms_bar.png")`.

Graph (Optional):
(Describe the graph and mention the graphs here)
(Insert graph here if applicable else remove the Graph heading and the graph info)

Conclusion:
(Use this space to write the conclusion for the scenario in 4–6 lines.)----------

Scenario 4: Aggregation + Pie Chart + Save (□ Medium)

Description:
(Use this space to write a description of the scenario in 4–6 lines.)----------

Task	Description
1.	Count the total number of games per `genre`.
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using `genre` as labels and `count` as values.
5.	Add percentage labels using `autopct` (e.g., '%.1f%%').
6.	Save the graph: `plt.savefig("genre_distribution.png")`.

Graph (Optional):
(Describe the graph and mention the graphs here)
(Insert graph here if applicable else remove the Graph heading and the graph info)

Conclusion:
(Use this space to write the conclusion for the scenario in 4–6 lines.)----------

Scenario 5: Advanced Analysis + Multiple Graphs (□ Hard)

Description:
(Use this space to write a description of the scenario in 4–6 lines.)----------

Task	Description
1.	Create a new column `score_category`: "Excellent" for score ≥ 9, "Good" for $7 \le$ score < 9, and "Average" for score < 7.
2.	Convert the `editors_choice` column: 'Y' $\rightarrow 1$, 'N' $\rightarrow 0$.
3.	Use NumPy's `np.diff()` to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4.	Plot a **Line Graph** showing the trend of average score per `release_year`.
5.	Plot a **Stacked Bar Chart** showing the count of `score_category` per `release_year`.

6.	Plot a **Histogram** showing the distribution of `score`.
7.	Save the graphs: `plt.savefig("score_trend.png")`, `plt.savefig("score_category_stacked.png")`, `plt.savefig("score_distribution.png")`.
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if `editors_choice` correlates with high scores.

Graph (Optional):
(Describe the graph and mention the graphs here)
(Insert graph here if applicable else remove the Graph heading and the graph info)

Conclusion:
(Use this space to write the conclusion for the scenario in 4–6 lines.)----------
ded into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Final Conclusion:

(Use this space to write the final conclusion for the scenario in 4–6 lines, which covers all the scenarios)

What you have learned

(Use this space to write the things that you have learned by doing this Project-1 for all scenarios in your own words.)